

Virtual Stakeholder System

Group #28

Project Deliverable #2

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1. Introduction

1.1. Document Objectives

The purpose of this document is to provide a comprehensive overview of the Virtual Stakeholder System, specifically outlining important aspects such as: the domain of the project, the main stakeholders and their goals, user stories, and a revised update of the preliminary requirements that were documented in the first deliverable. This document serves as a guide for understanding the Virtual Stakeholder System, its functionalities, expected interactions, and potential risks, which aid in the development and validation process.

1.2. Glossary

Table 1.1: Glossary

Term	Definition
GUI	Graphics User Interface
IEEE	Institute of Electrical and Electronics Engineers
LLM	Large Language Model
NF	Non-Functional Requirement
RE	Requirement Engineer
VSS	Virtual Stakeholder System

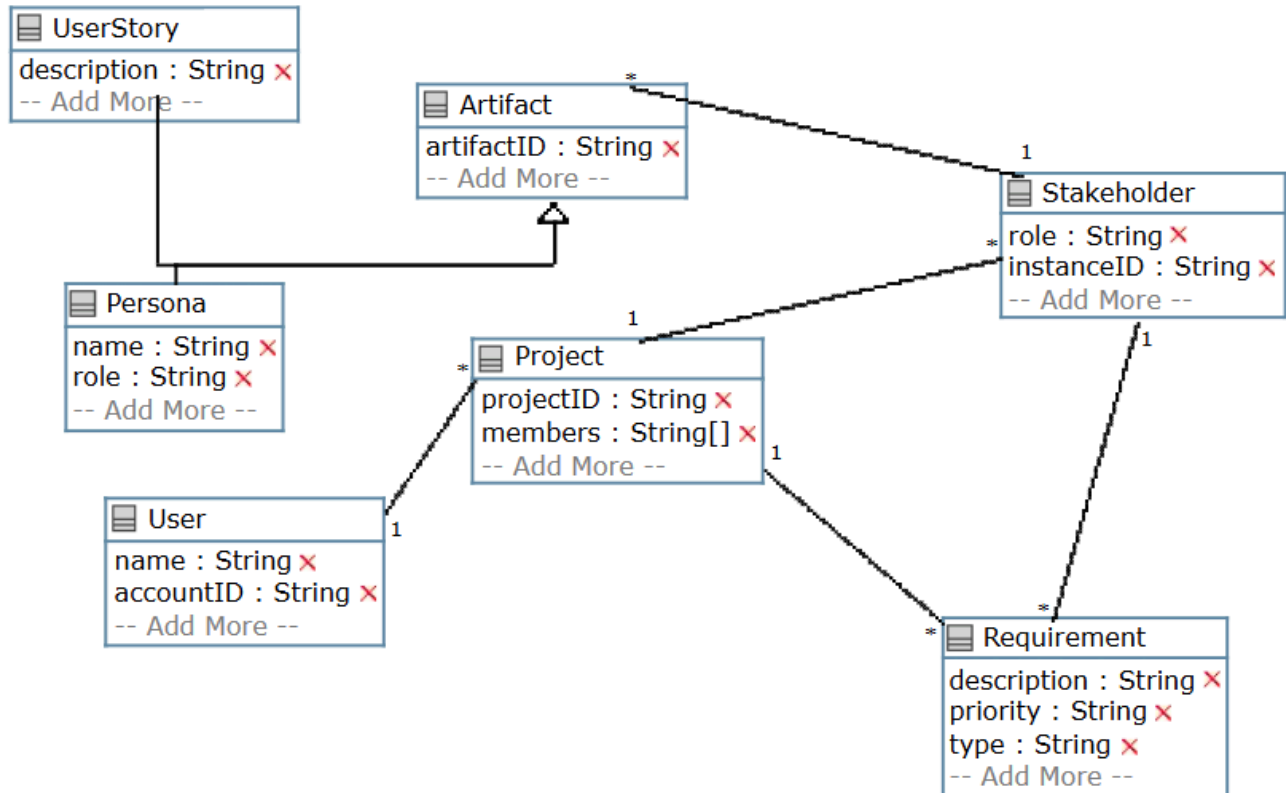
1.3. References

The main references in this document were the lecture notes presented in class, the content covered in the tutorials and labs, and the project description provided by the professor. A secondary resource we consulted was the IEEE 2018 standards document.

2. Domain Description

Domain Model (UML class diagram, without operations)

Figure 2.1: Domain Model



3. Initial Problem Description

3.1. Stakeholders and Their Goals

1. At least 5 important stakeholders, at least 1 goal per stakeholder and at least 12 goals in total (with unique identification)

Table 3.1: Stakeholders

Stakeholder	Goals
1. Requirements Engineer	G1: Efficiently gather and validate requirements from the system G2: Generate accurate project artifacts (personas, user stories, use cases) G3: Ensure requirements coherence and avoid duplicates
2. Project Manager	G4: Track requirement completion status and deadlines G5: Maintain project data security and confidentiality
3. Development Team	G6: Ensure clarity of generated non-functional requirements G7: Interact with virtual stakeholders for requirement clarification G8: Validate requirements against project goals
4. System Administrator	G9: Ensure system availability and performance G10: Maintain data security during storage and transmission G11: Support Multilingual system capabilities, specifically French & English G12: Regular system testing and updates
5. Non-Technical Users (Business Analysts, Product Owners)	G13: Interact with virtual stakeholders in preferred language G14: Access and understand generated artifacts easily G15: Provide domain specific feedback for stakeholder training G16: Save and Revisit stakeholder conversations
Total #: 5	Total #: 16

3.2 Scope of the system

1. General features within the scope of the system, and explicit description of what is not in the scope of the system
 - a. General Features within the scope of the system:
 - i. The VSS will be able to answer questions posed by the user, filling the role of the stakeholder.
 - ii. The VSS will allow users to create and manage personas based on LLM technology.
 - iii. The VSS will be able to critique requirements submitted by the user as well as provide optimal feedback.
 - b. Description of what is not in the scope of the system:
 - i. The VSS will not allow two or more collaborators to work together simultaneously in the same stakeholder instance.
 - ii. The VSS will not have a voice chat feature.
2. Description of dependencies and anticipated interactions with external systems

Table 3.2: Dependencies

Dependencies	Anticipated interactions with external systems
LLM - Required for generating stakeholder responses - Required for artifact creation and requirement validation - Needs regular updates and training	- API calls for stakeholder conversation responses - Requests for requirement validation and analysis
Database Management System - Required for storing project data, conversations and artifacts - Must support encrypted data storage and needs backup & recovery capabilities	- Real time data storage and retrieval
Authentication System - Required for user management and access control - Must support secure login and session management	- User authentication verification

3.3. Main User Stories

1. Develop at least 10 user stories (with identifiers) for your main features.
Please include at least one negative user story.

Table 3.3: User Stories

User Story 1: Project Manager	As a project manager, I want to make sure the needs of various stakeholders are elicited and met, in order to adjust project planning accordingly.
User Story 2: Project Team Member	As a Project team member, I want to ask the VSS for feedback on specific tasks so I can better prioritize my work.
User Story 3: Student	As a university student, I would like to use this system to assist me with writing valid requirements so that I can better learn the basics of software engineering.
User Story 4: Quality Assurance	As a quality assurance team member, I want to review current stakeholder feedback that has already been gathered, and verify that the given information is useful in order to improve the reliability of the system being built.
User Story 5: Cyber Security Specialist	As a Cyber Security Specialist, the VSS system should not compromise on any personal user data to keep the stakeholder's identities safe.
User Story 6: Hacker	As a hacker, I want to exploit unencrypted data in the system so that I can gain unauthorized access to stakeholder personal info.
User Story 7: Professor	As a Professor, I want a way to generate different personas to use as examples in my class and in exams, to help lessen my workload when building curriculums.
User Story 8: Technical Advisor	As a Technical Advisor, I want the system to provide validated responses and updated information in the technical domain.
User Story 9: Team Lead Coordinator	As a team lead coordinator, I want the system to produce requirement deadlines based on the project outline input by the user.
User Story 10: Software Developer	As a software developer, I want the system to produce use case scenarios that can help determine the end goals for the system.

3.4 Preliminary Requirements

1. Updated table of preliminary requirements following the interview and other sources of information (at least 35 requirements, including at least 7 non-functional requirements)

Legend: Blue = Modified Red = Removed Green = New

Table 3.4: Preliminary Requirements

ID	Requirement	Guidelines/ Notes	Priority (High, Medium, Low)	Type	Goal Identifier	User Story Identifier
1	The system will allow users to specify goals for their project.	On user request	High	F	G1, G8	
2	The system will automatically save stakeholder instances between sessions.		High	F	G16	
3	The system will allow users to save up to 15 stakeholder instances.		Medium	NF	G16	
4	The system will fully support bilingual functionality, allowing users to input queries and receive responses in either English or French.		High	NF	G11, G13	
5	The system will allow users to ask	By training a large	High	F	G1, G7, G13	User Story 2

	the virtual stakeholder questions.	language model				
6	The system will provide answers to users' questions.	Acting as the virtual stakeholder	High	F	G1, G7, G13	User Story 2
7	The system will allow users to create new stakeholder personas.	By training a large language model	High	F	G2	User Story 7
8	The system shall allow users to specify characteristics of created personas.	On user request	Medium	F	G2	User Story 7
9	The system will automatically produce persona artifacts.	On user request	High	F	G2, G14	User Story 7
10	The system will automatically update persona artifacts throughout the development process.	Through data gathered by the LLM	High	F	G2	User Story 7
11	The system will automatically produce user stories.	On user request	High	F	G2, G14	
12	The system will automatically produce non-functional requirements.	On user request	High	F	G6	User Story 3
13	The system will automatically produce use case	On user request	High	F	G2, G14	User Story 10

	scenarios.					
14	The system should produce requirement deadlines based on the project outline input by the user.	Deadlines should include time (24 hours), day, month and year	Medium	F	G4	User Story 9
15	The system must be able to validate certain requirements as specified by the user.	On user request	High	F	G1, G8	User Story 4 User Story 1
16	The system shall automatically detect duplicate and highly similar requirements and notify users of them.	By training a large language model on textual analysis and contextual data	High	F	G3	User Story 1
17	The system shall automatically detect requirements which are not coherent with each other and notify users of them.	By training a large language model on textual analysis and contextual data	Medium	F	G3	User Story 1
18	The system will incorporate human feedback for validation of individual or collective requirements.	By allowing users to leave feedback on requirements	Medium	F	G15	User Story 1 User Story 8
19	The system shall ensure all information on requirements and	Using industry standard security	High	NF	G5, G10	User Story 5 User Story 6

	artifacts are encrypted during storage and transmission of data.	protocols				
20	The large language model will not be trained on any personal user data to keep the stakeholder's identities safe.	Data includes: full name, address, email address and phone number	Low	NF	G5, G10	User Story 5 User Story 6
21	The system's virtual stakeholder must generate a response to a user's question in no more than 15 seconds.	On user request	High	NF	G7, G9	
22	The system shall produce requested artifacts in no more than 15 seconds.	On user request	High	NF	G9	
23	The system shall have 99.9% uptime.		High	NF	G9	
24	The system will maintain an error rate of no more than 0.1% for user requests.		High	NF	G3	
26	The system will be tested regularly to mitigate data changes and update responses based on newly identified benchmarks		High	NF	G12	

2. Ensure that all your goals and user stories are covered by at least one requirement.

- As seen in the goal identifier and user story identifier columns, each goal and user stories is relevant to at least 1 requirement, and in many cases a few.

3. Indicate how many requirements:

a) Were added compared to deliverable 1

Seven additional requirements were added to deliverable 1's total of 29 bringing the total to 36.

b) Were removed compared to deliverable 1

We felt #27 was redundant, so we removed it and incorporated the main idea into #28, bringing the total to 35.

c) Were modified compared to deliverable 1

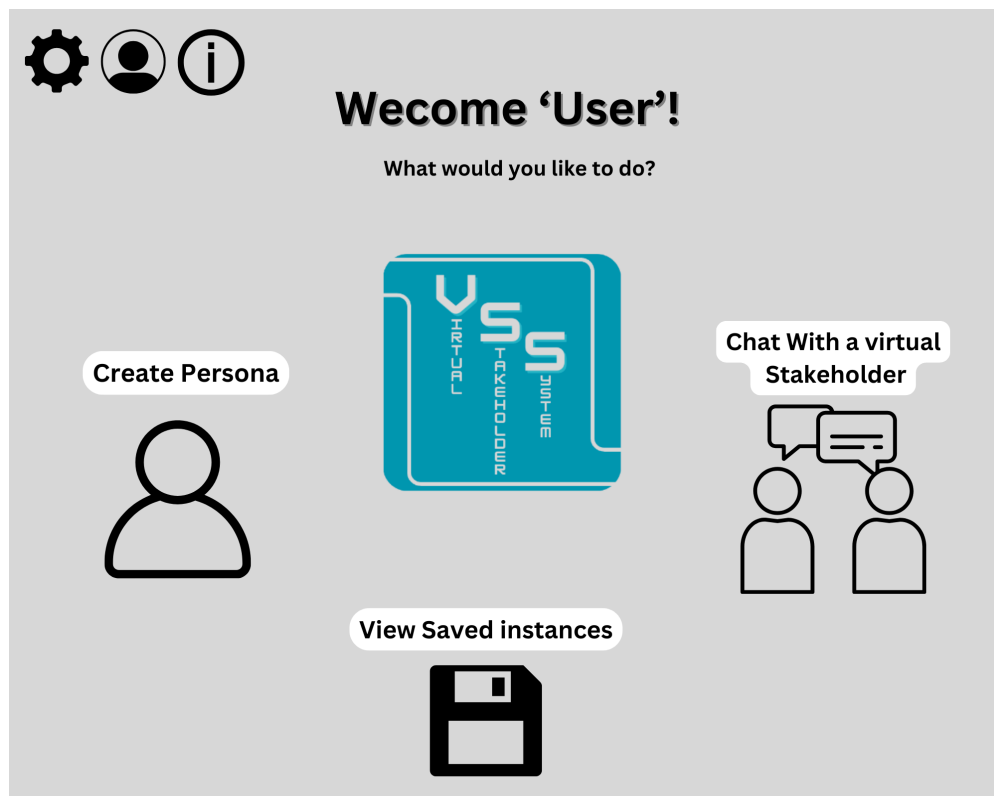
We modified two requirements in deliverable 2 - #4 and #28, to better describe what the requirements were trying to accomplish.

3.5 Risks

1. Security Risk: Hackers can gain access to the backend system and sensitive data, if the system does not address every possible security loophole.
2. Unvalidated outputs : The LLM's ability to provide a validated and updated response to the user's request can be compromised if the system is not updated regularly or the responses are not verified by the human stakeholders.
3. Malicious injection: Hackers can deliberately hack into the system to manipulate the LLM's responses.

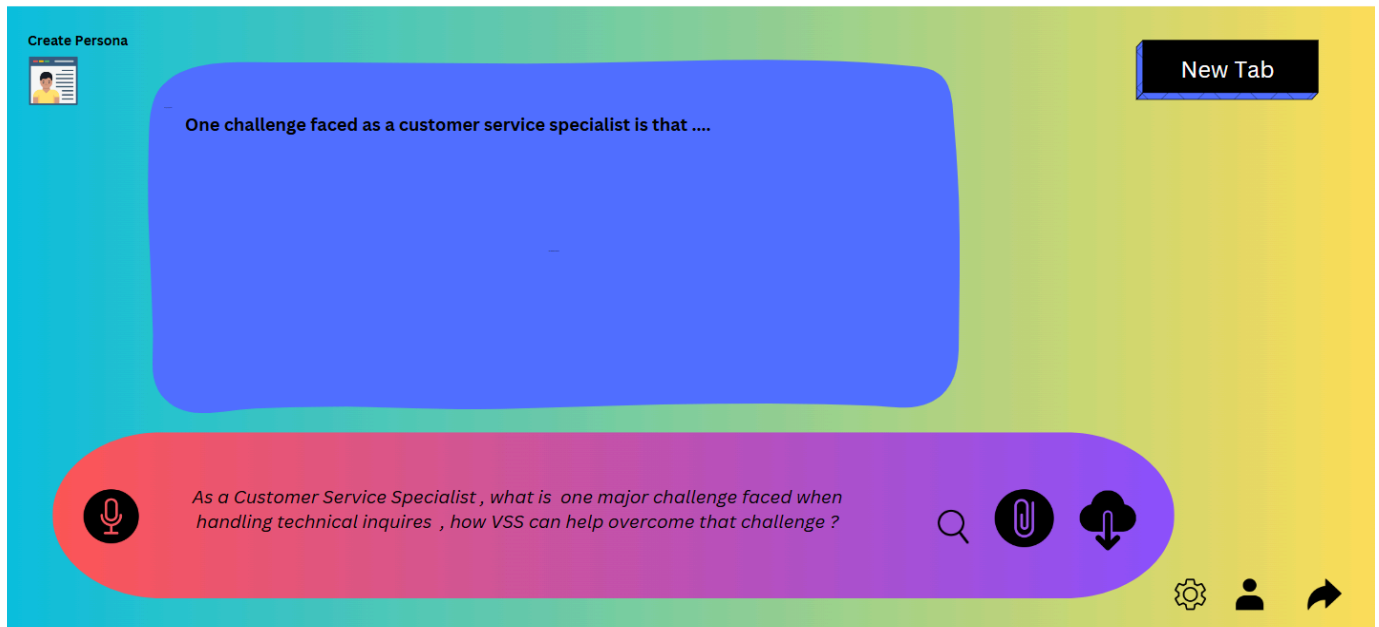
4. Interface Prototypes

Figure 4.1: Interface Prototype



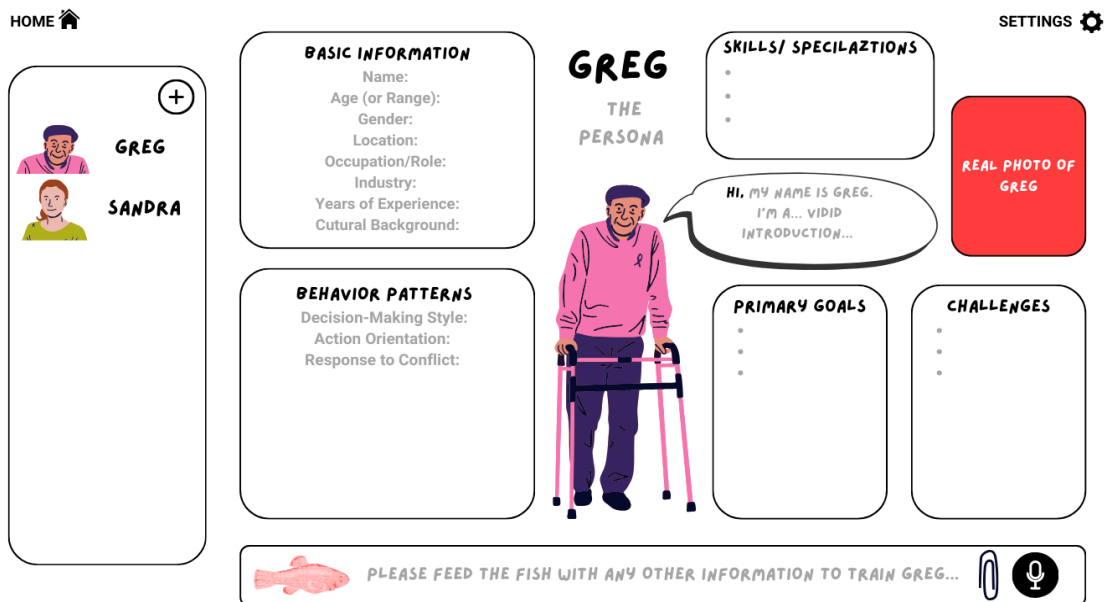
The above prototype provides insight into what a possible dashboard/homepage would look like within the VSS. There are options to create a persona, chat with a virtual stakeholder, and view previously saved instances. There are also settings, profile, and help buttons as well.

Figure 4.2: Chat Bot Prototype



This prototype presents a possible interaction with the virtual stakeholder chat bot. You can type/talk into the prompt, and are met with a response above, much like other common chat rooms/text platforms.

Figure 4.3: Persona



The above prototype shows what the persona creation tool could look like within the VSS. It displays all information in a concise and readable way, allowing users to inspect each aspect of the persona.

Team Description and Roles:

Who They Are?	Tasks	Percent
Anna	- Document Creation, Formatting and forwarding - Requirements - User Stories - Interface Prototype	20%
Alex	- Requirements - Introduction - Logo and Page Stylization - Formatting	20%
Caleb	- Requirements - User Stories - Scope - UML Diagram	20%
Owen	- Stakeholders and their goals - Scope - Requirements - Formating	20%
Mukhveer	- User Stories - Grammar Check - Formating - Interface Prototype - Risks	20%